

AI x Blockchain

Scam or Double Scam?

by Minh Huu Nguyen



DISCLAIMER

This seminar will focus solely on the technical aspects at the intersection of AI and blockchain.

Topics related to finance or fraudulent activities will not be covered.

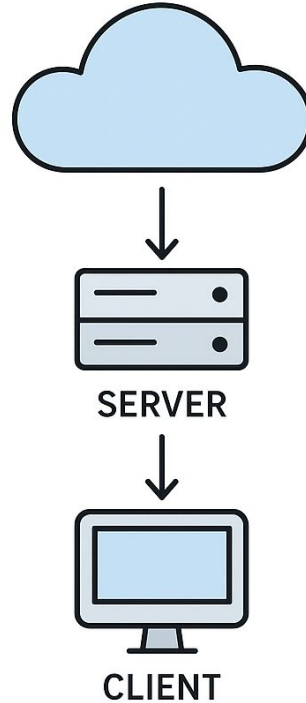
GENERAL CONCEPT



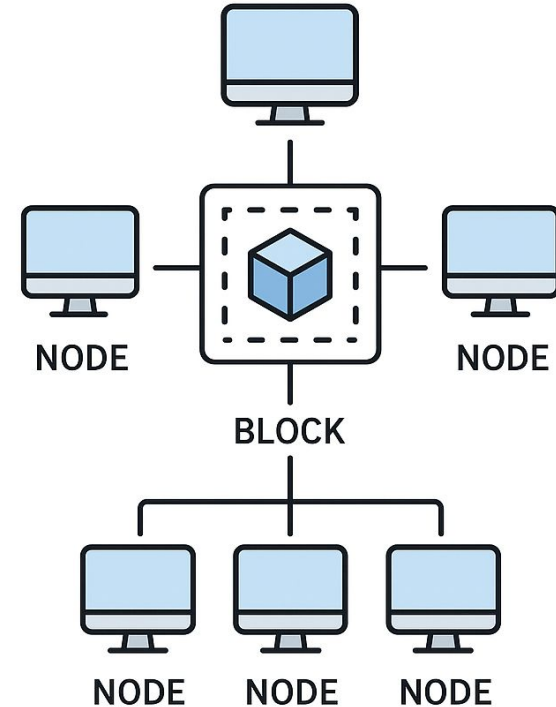
GENERAL CONCEPT

- All computations are executed by the **network's participating nodes**.
- **Regular users** (e.g. submitting transactions) do **not run computations directly**; they only submit requests to the network.
- Users who own nodes that **participate directly in block creation or provide services** can only **receive token rewards** if their contribution can be measured.

CLOUD COMPUTING



BLOCKCHAIN COMPUTING



ETH Example



Full Node: Stores the entire blockchain and validates all transactions and blocks according to consensus rules.

Archival Node: An “enhanced full node” that not only stores the current state but also retains the full historical state for every block.

Pruned Node (Trimmed Full Node): Stores all blocks but only keeps the current state and the necessary data to validate new blocks.

Light Node: Downloads only block headers and Merkle proofs to verify transactions.

Validator Node (Ethereum 2.0 / Proof-of-Stake): In Ethereum PoS, validators participate in creating new blocks and confirming existing blocks.

Node Type	Blockchain Storage	Current State	Historical Access	Load / Weight
Full Node	✔ Yes	✔ Yes	✘ No (current only)	Medium
Archival Node	✔ Yes	✔ Yes	✔ Yes (full history)	Very Heavy
Pruned Node	✔ Yes	✔ Yes	✘ No (pruned)	Lighter than Full
Light Node	✘ Only headers	✔ Yes	✘ No	Light
Validator Node	✔ Yes	✔ Yes	✘/✔ Depends on config	Medium

COOL PROJECTS

AI x BLOCKCHAIN



Cortex - CTXC Token

Targets: Extends blockchain capabilities by integrating machine learning models into smart contracts, which cannot be done in original ETH blockchain.

Techniques:

- **Cortex Virtual Machine (CVM):** similar to EVM, but it can run matrix calculation efficiently so AI developer and upload AI model to Cortex blockchain to run inference.
- **Proof-of-work:** to motivate miner run on GPU



Render Network - RENDER Token



Targets:

- Build a decentralized global GPU infrastructure for both graphics rendering and AI workloads by leveraging idle GPU resources.
- Reduces computation costs for users compared to centralized cloud services.
- Enabling GPU owners to monetize their excess capacity.

Render Network - RENDER Token



Techniques:

- **Decentralized Peer-to-Peer GPU Marketplace:** marketplace to match "creators" (who need GPU) and "node operators" (who provide GPU).
- **Tokenomics:** creators pay token, node operators stake token to show commitment. After finishing job, node operators will be paid reward token.
- **Proof-of-Render:** technique to validate the results from node operators before paying reward token.
- **Smart Contracts & Blockchain Layer:** automate the process (job distribution and payment) by using blockchain (ETH before, SOL now).
- **Digital Rights Management:** data is encoded and split before sending to nodes. Node operators cannot access the raw data.

The techniques behind are quite cool
but they are not AI... So we skip....

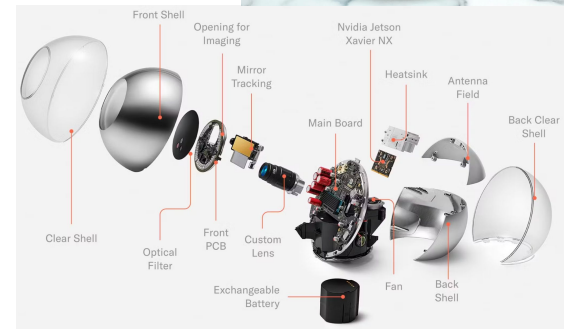
Worldcoin – World / WLD Token

Co-founder by Sam Altman  OpenAI

Targets: Building a decentralized global identity system called World ID to distinguish real humans from AI bots on the internet.

Techniques:

- **Orb and IrisCode:** A tool to scan human eyes and a pipeline to encode the eyes into IrisCode (binary code).
- **Secure Multi-Party Computation:** Split the code into shares and save on multiple parties. Encrypt the code into a type which cannot be reversely decoded back into the raw data.
- **Zero-Knowledge Proofs and World App:** In production, instead of use the eyes to verify, user use the app (which contains private key). The app will create a ZK-proof to verify.



Worldcoin – World / WLD Token

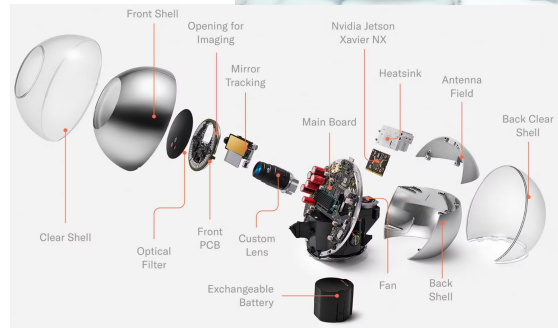
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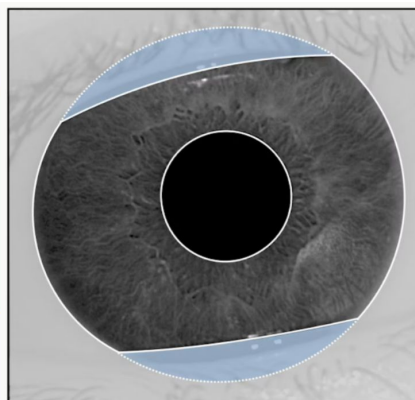
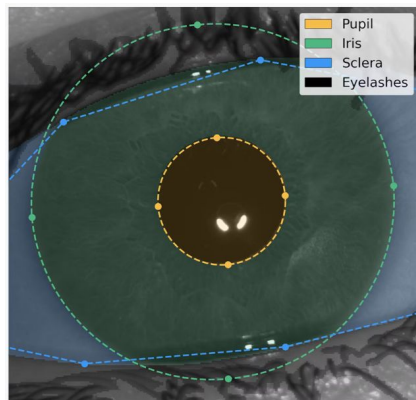
WHERE IS AI?



Worldcoin - World / WLD Token



- Liveness detection and anti-spoofing detection to avoid invalid input data, on edge device.
- Image processing and feature extraction to get the IrisCode, on edge device.
- Vector search algorithms (Approximate Nearest Neighbors with indexing FAISS, HNSW) for fast vector query.



raw iris texture



iris code

Alethea AI - ALI Token

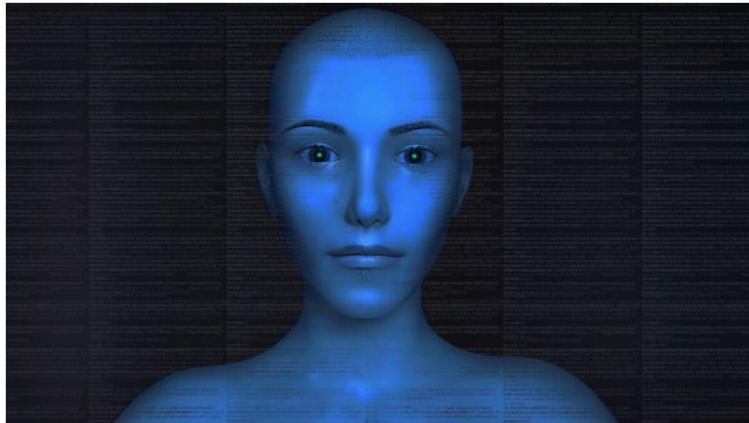
ALETHIA AI



Targets: AI + NFTs = Intelligent NFT (iNFT). iNFT can be animated, interactive, and capable of learning.

Techniques:

- **iNFT** = NFTs + AI = **NFT base + Personality Pod** (AI model for conversation, language and animation)
- **CharacterGPT**: prompt-to-3D-character GenAI model

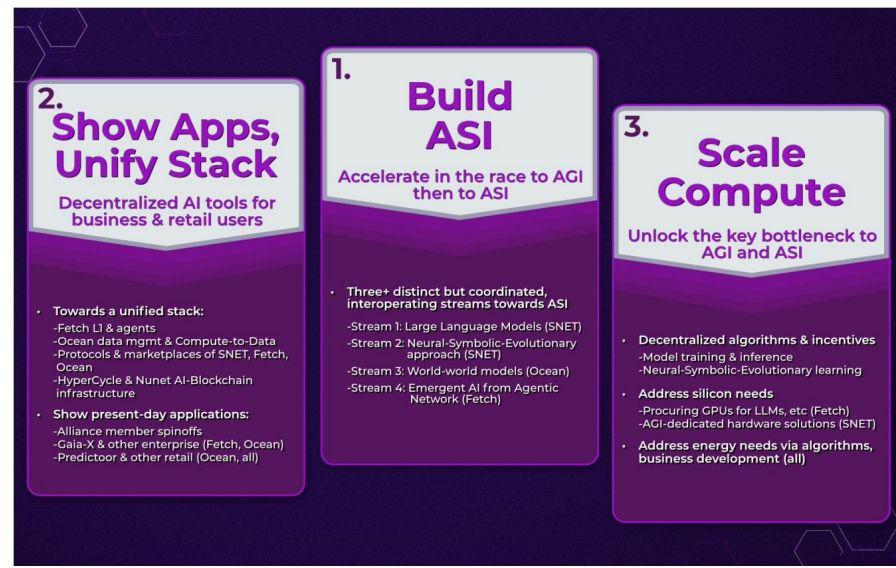


Artificial SuperIntelligence Alliance - ASI Token



Build ASI:

- Large Language Models - LLMs
- Neural-Symbolic Evolutionary Approach: OpenCog
Hyperon - a platform to unify various inference systems.
- World-World Models: Continuously use physics information from real-world to to train the model (??)
- Emergent AI from Agent Networks: Build a network which contains lots of simple AI Agents, and then combine them to create ASI.



Artificial SuperIntelligence Alliance – ASI Token



Show Apps, Unify Stack:

- Decentralized AI Agent and AI Agent interaction protocol (like MCP)
- Data tokenization and Compute-to-Data to ensure data privacy.

Scale Decentralized Compute:

- Develop decentralized algorithms to run on blockchain.
- Motivate community to join to contribute the hardware and get reward.

